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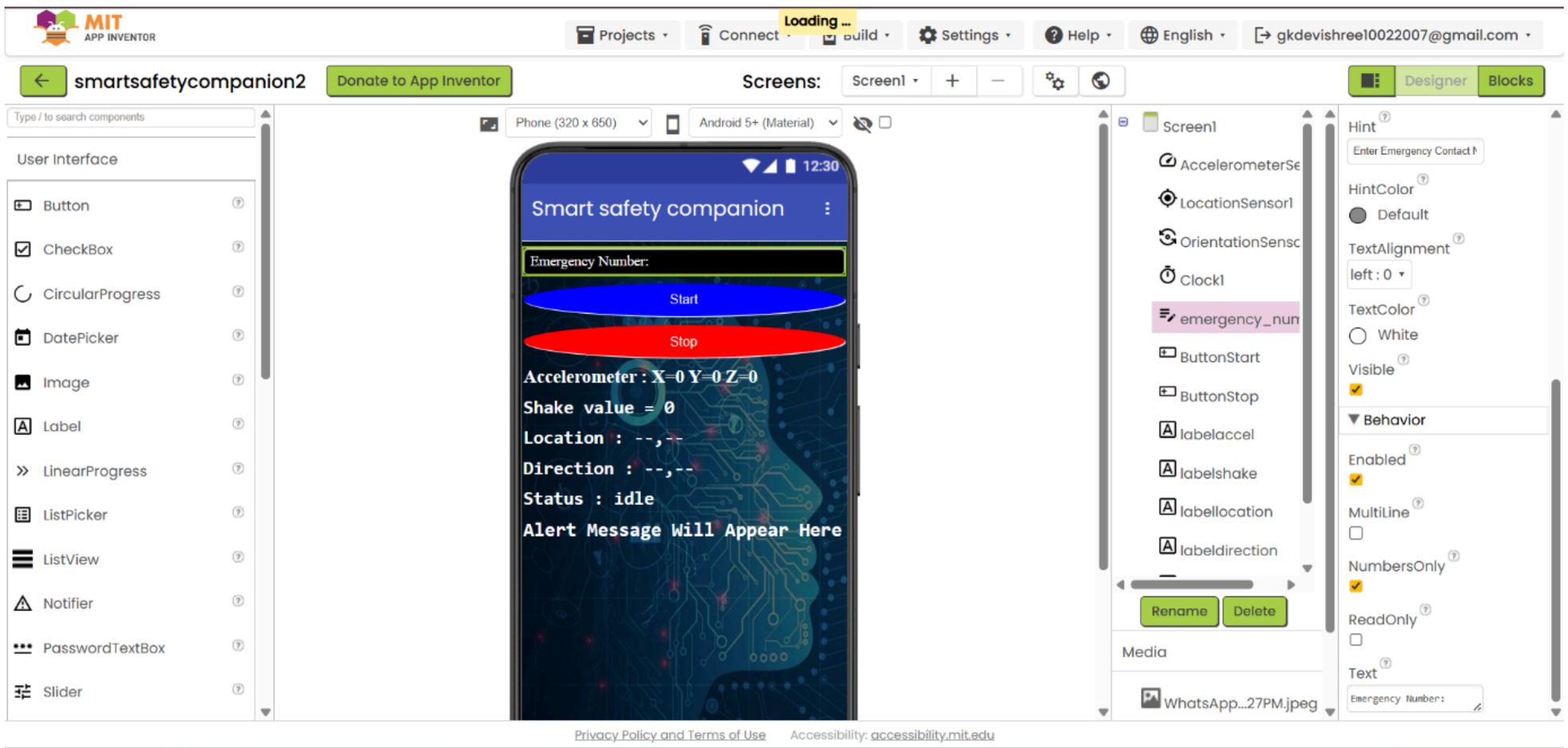
PROTOTYPE EXTENSION

Extended User Interface Design of Smart Safety Companion using MIT App Inventor



This section presents the extended user interface of the Smart Safety Companion application developed using MIT App Inventor. The interface has been enhanced by adding new interactive components such as monitoring control buttons, sensor information labels, and emergency alert sections.

System Operation Description

This section describes the overall working of the proposed system. The application continuously collects data from built-in sensors such as the accelerometer and GPS. The collected data is analyzed in real time to identify abnormal or emergency conditions. When such conditions are detected, the system responds by displaying alerts and relevant information on the user interface. This process ensures continuous monitoring, timely response, and improved personal safety. The interface is designed to be user-friendly and easy to navigate. Start and Stop buttons are provided to control safety monitoring efficiently.Real-time sensor data is displayed clearly for better understanding.

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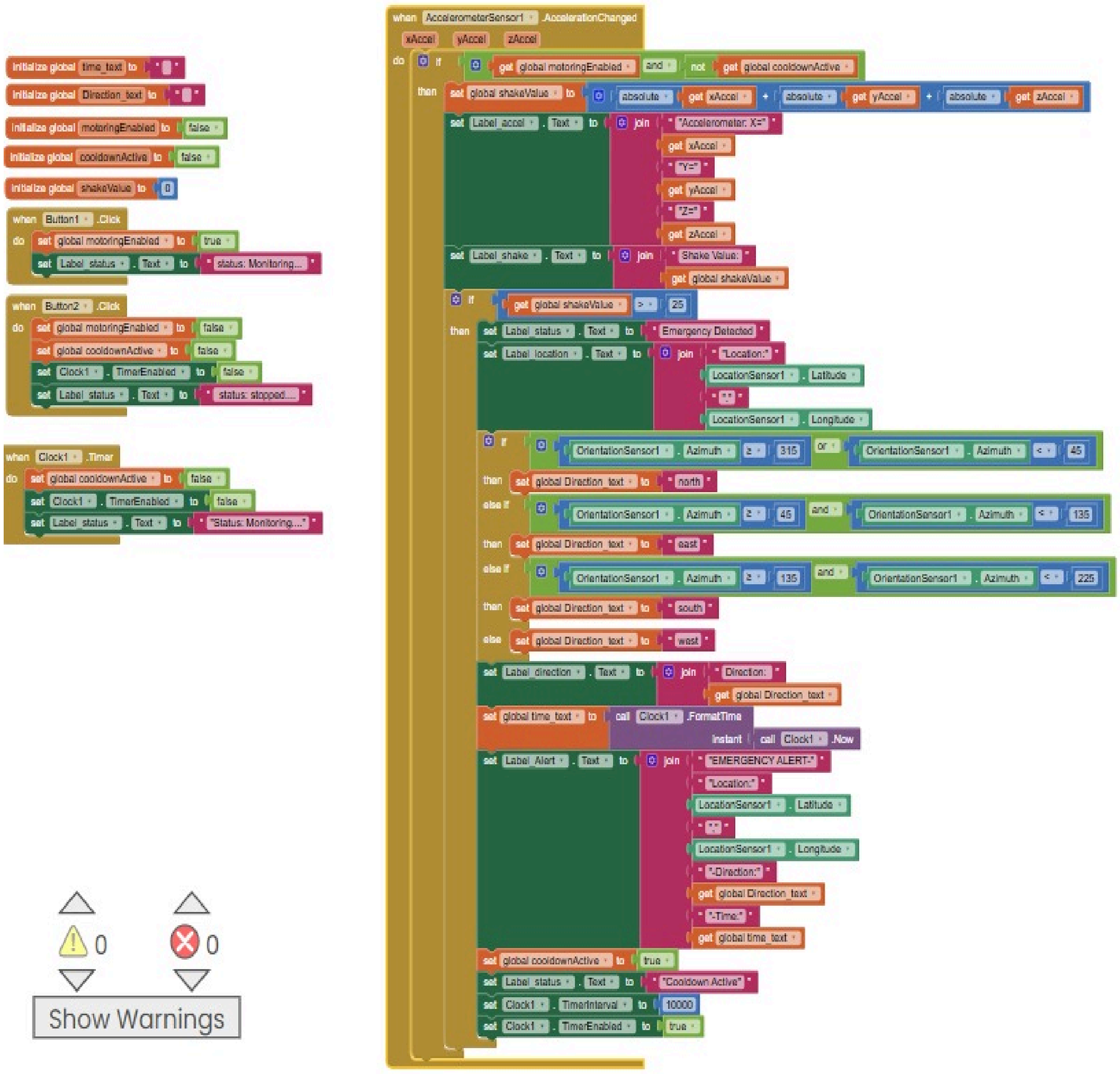
BLOCK LEVEL IMPLEMENTATION



Block Diagram Description:

The block diagram illustrates the extended prototype architecture, where sensor and location data are collected and processed in real time. Based on the analyzed data, the system identifies emergency conditions and triggers alert mechanisms to improve safety and responsiveness.

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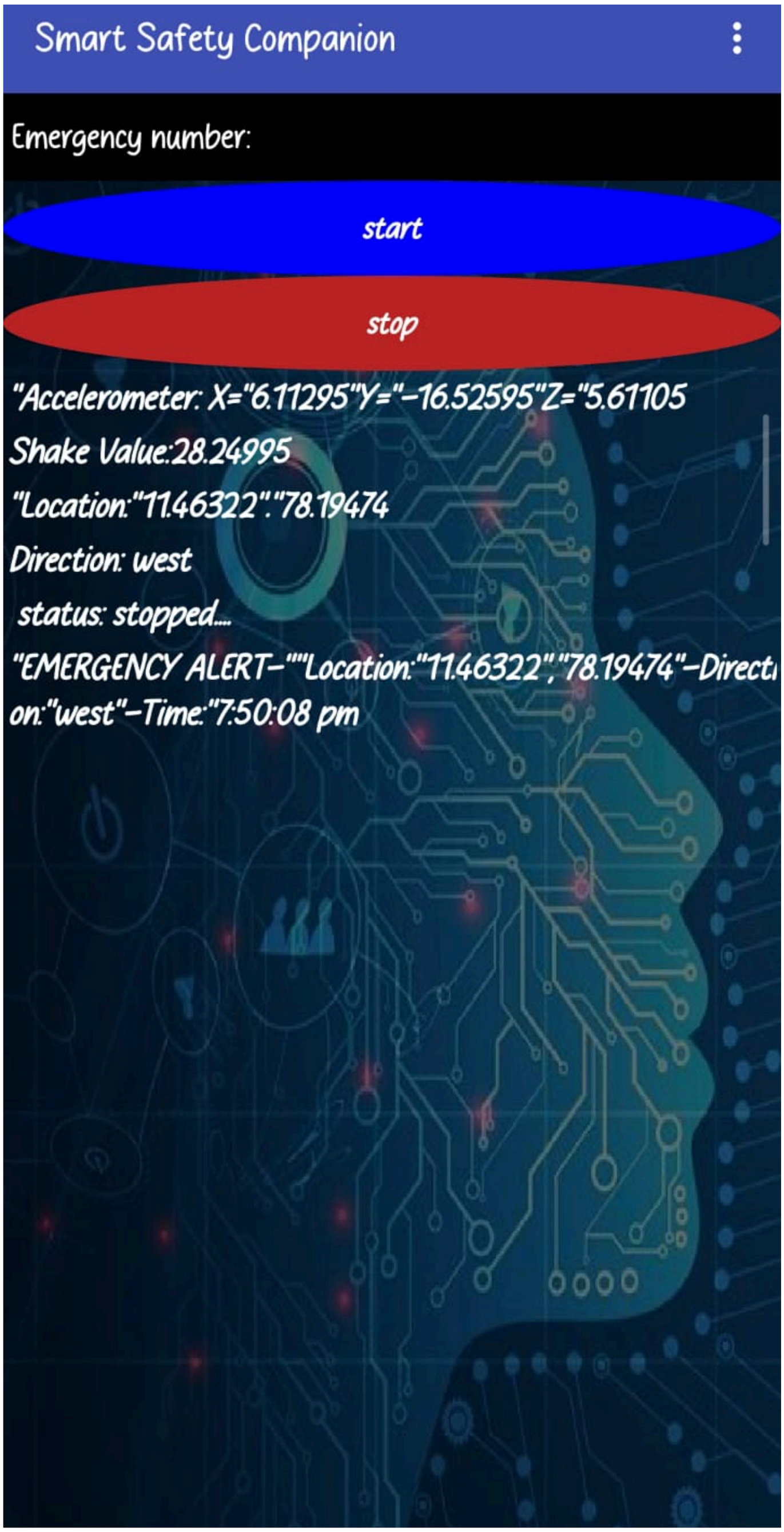
OUTPUT AND RESULT



The output of the prototype extension demonstrates the effective functioning of the proposed system in real-time conditions. The application continuously displays live sensor readings, including accelerometer values, which are used to monitor user

movement and detect abnormal patterns. Along with sensor data, the system shows real-time location and direction information obtained through the GPS module, enabling accurate user tracking. When the system identifies emergency or abnormal conditions based on predefined thresholds, it immediately triggers alert mechanisms. These alerts are clearly displayed on the screen. The system successfully detects emergency situations, process data efficiently.

CONCLUSION

The screenshots demonstrate the successful extension of the original prototype by integrating real-time sensor monitoring, emergency detection, GPS-based location tracking, and automated alert generation. These improvements significantly enhance the reliability, efficiency, and real-world applicability of the Smart Safety Companion.